Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER B.TECH. (ELECTRICAL &ELECTRONICS)

MAKE-UP EXAMINATIONS, DECEMBER 2024 - JANUARY 2025

DIGITAL SIGNAL PROCESSING [ELE 3122]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 30 December, 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Determine y[n] in time-domain ***(04)***

$y(n) = u[n-1] * u[n-3]$

1B. ***(02)***

Determine the signal $x[n]$ if the DTFS representation of the signal x(n) with period N =3 is written as

$$x(n) = \sum_{k=0}^{N-1} X[k] e^{j(\frac{2\pi nk}{N})}$$

The discrete-time Fourier series coefficient as X[-3] = 2, X[4] = 1 and other coefficient is zero.

1C. ***(04)***

Demonstrate the overlap-save method to compute the linear convolution of the following sequences. $h[n] = [\underset{\uparrow}{1}, 1, 1]$, and

$x[n] = [\underset{\uparrow}{1}, 2, 0, -3, 4, 2, -1, 1, -2, 3, 2, 1]$. Assume sub-frame length =5.

2A. Determine the signal x(n) using IDTFT for given ***(03)***

$X[e^{j\omega}] = e^{-j\omega}[1 + \cos(w)]$

2B. ***(03)***

For the DFT sequence $X[k] = [\underset{\uparrow}{2}, \ 2+3j, \ 0, \ 2-3j]$,

(i) Calculate the IDFT of the data sequence X[k]
(ii) Calculate the IDFT of the circularly right shifted X[k] by 2 units.

2C. Apply DIT-FFT radix-2 algorithm, to compute the 8-point DFT of the sequence $x[n] = [1,1,1,1,0,0,0,0]$. Draw the Butterfly diagram (signal flow graph). Show all the intermediate results to arrive at final answer on the Butterfly diagram. ***(04)***

3A It is required to optimize signal-to-noise ratio and sensitivity of certain digital system performing low-pass filtering and satisfying the linear phase constraint. Achieve it by designing a suitable fourth-order filter with a cut-off frequency of $\omega_c = \frac{\pi}{4}$. Use Hanning window. Also, draw the linear phase structure for the designed filter. ***(05)***

3B List any four differences between FIR and IIR filters ***(02)***

3C Using partial fraction expansion technique, find the inverse Z-transform of the following: ***(03)***

$X[Z] = \frac{z}{3z^2-4z+1}$; ROC $\frac{1}{3} < |z| < 1$

4A You are given with a simple signal $x[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 2\delta[n-3] + \delta[n-4]$. It is possible to obtain some information of the simple signal on their DTFTs. ***(03)***

(i) Find $X(e^{j0})$ and $X(e^{j\pi})$ without computing the DTFT of $X(e^{j\omega})$

(ii) Find $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

(iii) Is phase of $X(e^{j\omega})$ linear?

4B Obtain the Direct form-I and Direct form-II structure for the following system ***(03)***

$y[n] = 4y[n-1] - 2y[n-2] + x[n] + 3x[n-1]$. Why Direct form II structure is better than Direct form-I structure?

4C Derive the system function and frequency response equations for the DT system characterized by the pole – zero map shown in Figure Q4A. The poles are placed at a distance of r from the origin. Zeros are placed on unit circle. Draw the magnitude plot assuming the maximum magnitude is unity. Is the given system IIR or FIR system? Justify your answer. ***(04)***

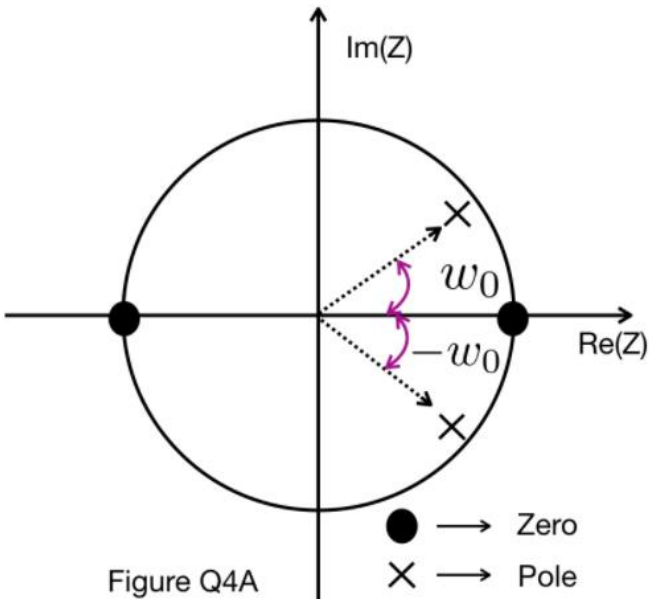


Figure Q4A

5A Let $x(t) = \cos(10\pi t) + \cos(30\pi t) + \sin(30\pi t)$ be sampled at 20 Hz and reconstructed using ideal low pass filter with a cut-off frequency of 20 Hz. ***(02)***

a. Find expressions for sampled and reconstructed signals

b. Determine the frequencies present in the reconstructed signal

5B Apply Impulse Invariance Technique to convert analog filter defined by the transfer function $H(s) = \frac{4s+11}{s^2+5s+6}$ to its equivalent digital filter. Assume sampling frequency of 10 Hz. ***(03)***

5C For a certain biomedical application, it is required to remove a band of high frequencies. Design a suitable Butterworth filter using Bilinear transformation technique to meet the following specifications: ***(05)***

$$0.85 \leq \left|H\left(e^{j\omega}\right)\right| \leq 1 \;;\quad 0 \leq \omega \leq 0.4\pi$$

$$\left|H\left(e^{j\omega}\right)\right| \leq 0.3 \;;\quad 0.75\pi \leq \omega \leq \pi$$

Take T = 1 sec